TEXORA · LECTURE SHEET

Quality Calculation

The numbers that measure quality: DHU for in-process control, the ISO 2859-1 AQL chain for shipment decisions, rejection rates, and the four-point system for fabric inspection — each with a fully worked decision.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 9 · SPECIALIZATION — APPAREL MANUFACTURING

1. DHU — defects per hundred units

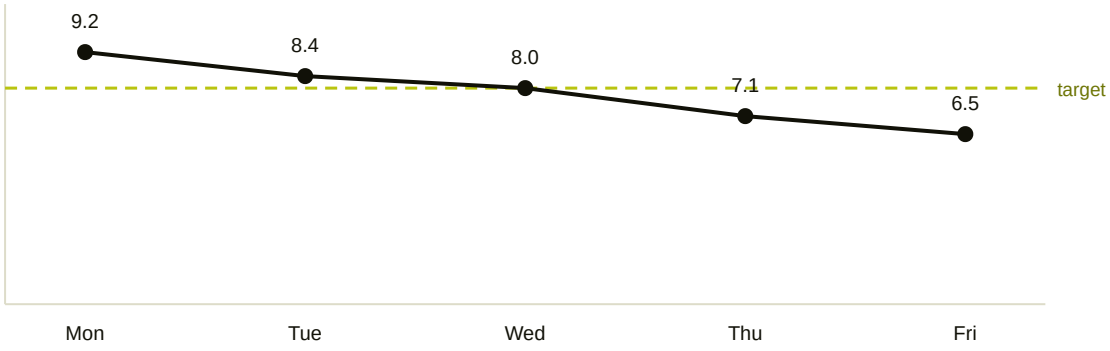
In-process quality is steered by **DHU**, the defects found per hundred units inspected at a checking point:

$$\text{DHU} = (\text{defects found} \div \text{units inspected}) \times 100$$

WORKED EXAMPLE — A WEEK OF END-OF-LINE DHU

DAY	INSPECTED	DEFECTS	DHU
Mon	500	46	9.2
Tue	500	42	8.4
Wed	500	40	8.0
Thu	550	39	7.1
Fri	550	36	6.5

Friday's DHU = 36 ÷ 550 × 100 = 6.5. The week's trend — not any single day — is the management information: a new operator was trained on Tuesday and the dominant defect (skip stitch at the side seam) was traced to one machine on Wednesday; the falling line is the signature of those two fixes. A **spot check** on Friday found 68 defects in 850 garments checked across the day at a second point: DHU = 68 ÷ 850 × 100 = **8.0** — higher than the end-of-line figure, because it counts defects before the repair loop, which is exactly what an in-process figure should do.



End-of-line DHU across the week, against the factory target of 8.0

Figure 1 — DHU trend chart. Plotted daily at the same checking point with the same defect catalogue; the target line turns the chart into a decision, not a decoration.

2. The AQL decision — ISO 2859-1

Shipment decisions use acceptance sampling exactly as built in TE-3107: defect classes critical (zero tolerance), major (AQL 2.5) and minor (AQL 4.0), General Inspection Level II. The course order ships in lots of 2,400 shirts, so the chain is the established one: lot 1,201–3,200 → **code letter K** → sample **n = 125** → majors **Ac 7 / Re 8**, minors **Ac 10 / Re 11**.

> **WORKED EXAMPLE — AUDITING THE COURSE ORDER**
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> The order divides into 12,000 ÷ 2,400 = **5 lots**; total garments sampled across the order = 5 × 125 = **625**. In the first lot's audit the inspectors find 5 major-defective and 9 minor-defective garments, no criticals. Majors 5 ≤ 7 — pass. Minors 9 ≤ 10 — pass. **The lot ships.** Had minors reached 11, the lot would have failed on minors alone and gone to 100% re-inspection and repair before re-audit: one failing class fails the lot, however clean the others are.

3. Rejection rate and the cost of failure

After final inspection, the **rejection rate** = rejected garments ÷ garments inspected × 100 summarises what could not be repaired into first quality. In one lot of the course order, 36 of 2,400 garments were finally rejected (second quality or waste): 36 ÷ 2,400 × 100 = **1.5%**. Each rejected garment has already consumed its full fabric, trims and CM — at the Topic 6 cost of $4.27, one lot's rejections destroy 36 × 4.27 ≈ **$154** of finished value, before counting the inspection and repair labour that preceded the decision. Quality calculation exists to price prevention correctly against this figure.

4. Fabric inspection — the four-point system

Fabric quality is calculated before cutting by the **four-point system**: defects are penalised by size (up to 3 in: 1 point; 3–6 in: 2; 6–9 in: 3; over 9 in or a hole: 4), and the roll is graded by points per 100 square yards, with 28 points a common acceptance limit (the limit used in TE-3202).

> **WORKED EXAMPLE — GRADING A 100 M ROLL**
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> Roll: 100 m long × 1.50 m wide. Defects found: three ≤ 3 in (3 × 1 = 3 pts), two of 3–6 in (2 × 2 = 4 pts), one of 6–9 in (1 × 3 = 3 pts). Total = **10 points**.
>
> Fabric area = 100 × 1.50 = 150 m² = 179.4 yd². Points per 100 yd² = 10 × 100 ÷ 179.4 = **5.6** — comfortably inside the 28 limit: **the roll is accepted** and released to the cutting plan of Topic 3.

Glossary

TERM	MEANING
DHU	Defects per hundred units inspected; the in-process quality rate.
AQL	Acceptance Quality Limit (ISO 2859-1): worst tolerable process average, percent defective, for a series of lots.
Ac / Re	Accept and reject numbers of a sampling plan; Re = Ac + 1.
Rejection rate	Finally rejected garments as a percentage of garments inspected.
Four-point system	Fabric grading by penalty points per defect size, normalised to points per 100 yd².

Key points and summary

- DHU steers production daily: count defects, divide by units inspected, plot the trend against a target.
- AQL decides shipment statistically: course lots of 2,400 → code K → n = 125 → majors Ac 7/Re 8, minors Ac 10/Re 11 — the same chain as TE-3107.
- Rejection rate prices failure: 1.5% of a lot ≈ $154 of finished value destroyed.
- Fabric is graded before cutting by the four-point system (worked roll: 5.6 points/100 yd² against a limit of 28 — accept).

Practice

Answer in your own words; brief model answers follow.

1. A checker inspects 640 garments and finds 51 defects. Calculate the DHU and state whether it meets a target of 8.0.

2. In a lot audit (n = 125), inspectors find 7 major-defective and 11 minor-defective garments. Give the decision for each class and for the lot.
3. A 120 m roll, 1.45 m wide, collects 26 penalty points. Grade it per 100 yd² against the 28 limit. (1 m² = 1.196 yd².)
4. Why is a single day's DHU a weak basis for action compared with the week's trend?

Answer key

1. DHU = 51 ÷ 640 × 100 = 7.97 ≈ 8.0 — just meets the target.
2. Majors 7 ≤ Ac 7 — pass. Minors 11 ≥ Re 11 — fail. The lot fails on minors: 100% re-inspection and repair of the minor class, then re-audit.
3. Area = 120 × 1.45 = 174 m² = 208.1 yd². Points per 100 yd² = 26 × 100 ÷ 208.1 = 12.5 — within the limit; accept.
4. One day mixes sampling variation, style mix and inspector variation. A trend over days at the same checkpoint separates a real process change (which needs action) from noise (which does not).

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